

Data Preparation Session 6

Exercises

Exercise 6.1.

- a) Load the “dirty petal.csv” file.
- b) Search for outliers and propose a solution for them. Solve them with your proposal.
- c) Propose a solution for the NAs using the information in the dataset only. Implement your proposal. Attention: now there are more NAs than yesterday. You replaced outliers with NA, and they counter-attack. May the force with you.

Exercise 6.2.

- a) Load the “olympics.txt” file. Figure out the file type.
- b) The data is not tidy. Make it tidy by pivot transformations.
- c) Check for duplicates. Eliminate them.
- d) Check for outliers, propose a solution.
- e) Check the coherence between variable names and content, solve the problem.

Exercise 6.3.

1. Read the immunization data.
2. Check the tail of the dataset and remove the unnecessary rows.
3. Check for missing data and replace them by NA.
4. Rename columns and transform them to numeric (if possible):
 - ‘Time’ to year,
 - ‘Population, total [SP.POP.TOTL]’ to pop after division by 1,000,000
 - ‘Mortality rate, under-5 (per 1,000 live births) [SH.DYN.MORT]’ to mort
 - ‘Immunization, measles (% of children ages 12-23 months) [SH.IMM.MEAS]’ to imm
 - ‘GDP per capita, PPP (constant 2011 international \$) [NY.GDP.PCAP.PP.KD]’ to gdppc
5. Create two new numeric variables:
 - surv as $(1000 - \text{mort}) / 100$
 - lngdppc as the logarithm of gdppc
6. Rename two character-type variables:
 - ‘Country Name’ as countryname
 - ‘Country code’ as c
7. Rearrange the order of the variables in the following way: year, countryname, c, pop, mort, surv, imm, gdppc, lngdppc
8. Remove all observations before 1998 and all observations with NA in pop, mort and imm.
9. Store the number of unique country names in a scalar called “number” and the number of missing values in the gdppc column in “miss_in”. Print number and miss_in.
10. Create a dataset where one row is one year, and one column is one country, and the values are the values of the imm variable.